

Word Problems & Applications

Translating · Number · Money · Mixture · Distance–Rate–Time · Percent · Geometry

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Set up and solve word problems independently across all seven problem types.

Show your full working in the space provided under each question — begin with a 'let' statement and an equation. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the Answer Key.

Level 1 — Fluency

1. Write an expression for 'seven less than a number n '. **[Translate]**
2. The sum of a number and 15 is 40. Find the number. **[Number]**
3. Pencils cost \$3 each. How many can be bought with \$27? **[Money]**
4. Find 10% of \$250. **[Percent]**

5. A bus travels at 50 km/h for 4 hours. How far does it go? **[Distance]**

6. A square has perimeter 48 cm. Find one side length. **[Geometry]**

Level 2 — Application

7. Three times a number, decreased by 8, is 25. Find the number. **[Translate & solve]**

8. The sum of two consecutive integers is 55. Find them. **[Consecutive integers]**

9. A plumber charges a \$60 call-out fee plus \$45 per hour. A bill was \$240. How many hours of work? **[Wages]**
10. How many litres of pure water must be added to 2 L of 60% juice to make it 40% juice? **[Mixture]**
11. After 20% off, a coat costs \$72. What was the original price? **[Percent (work back)]**
12. Two angles are supplementary (sum to 180°). One is 40° more than the other. Find both. **[Angles]**

Level 3 — Thinking

13. Five times a number minus 3 equals twice the number plus 12. Find the number.

[Equation, both sides]

14. The sum of three consecutive even integers is 78. Find the largest. **[Consecutive even]**

15. A jar holds nickels and dimes: 20 coins worth \$1.65 in total. How many of each?

[Coins]

16. Tea worth \$5/kg is mixed with tea worth \$9/kg to make 12 kg of blend worth \$6/kg. How many kg of each? **[Blend cost]**
17. Two trains leave the same station in opposite directions at 80 km/h and 100 km/h. After how many hours are they 540 km apart? **[Closing distance]**
18. A rectangle's length is 4 cm more than twice its width. Its perimeter is 56 cm. Find the dimensions. **[Dimensions]**